

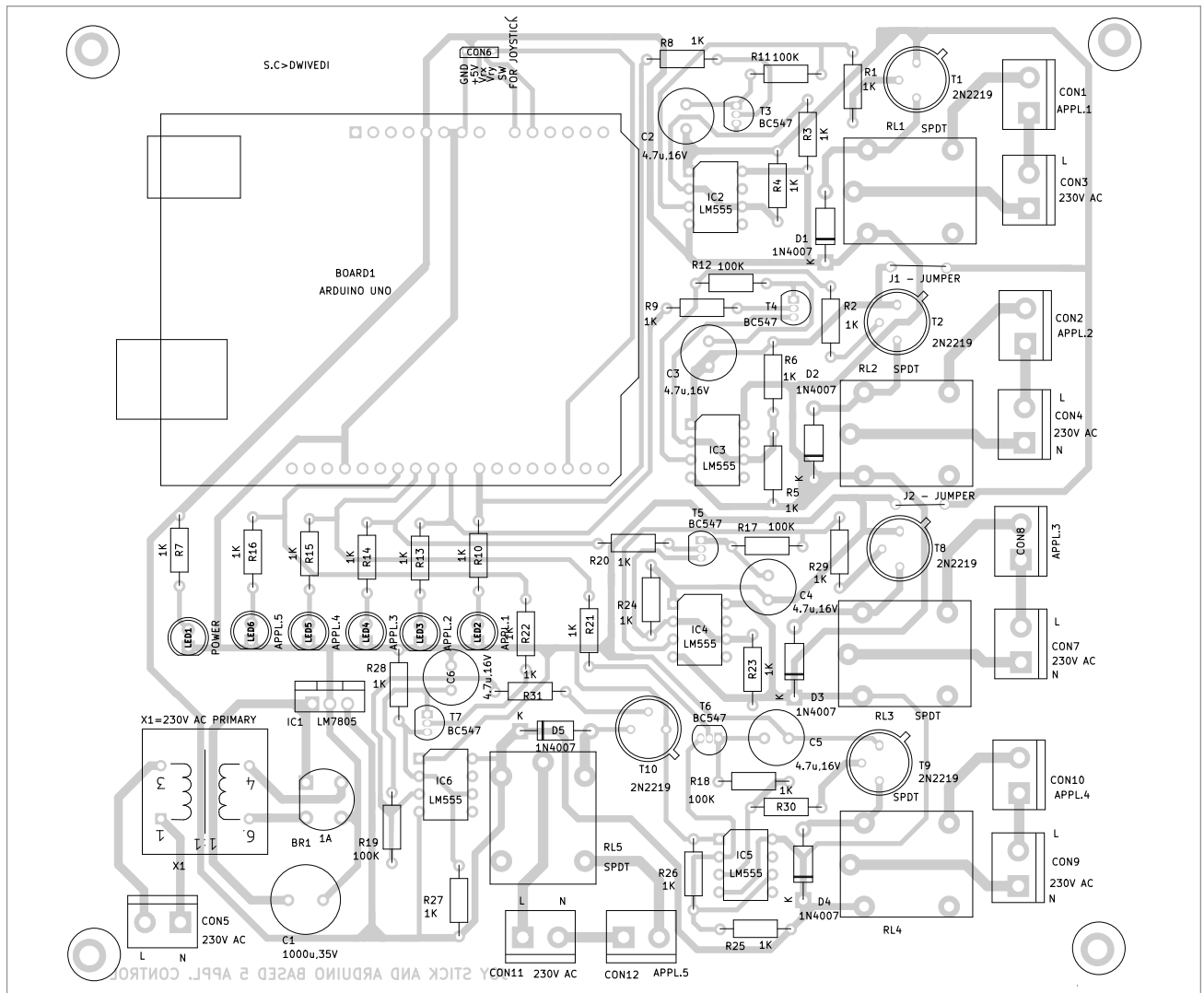


Fig. 6: Component layout of the PCB

You can momentarily turn on (blink) the five LEDs in each direction according to the joystick shaft's movement. An actual-size, single-side PCB for the project is shown in Fig. 5 and its component layout in Fig. 6.

Assemble the circuit on the PCB and enclose it in a cabinet. Connect all the connectors (CON1 through CON6) at the rear side of the cabinet and LEDs (LED1 through LED6) at the front side. LED1 indicates power, while LED2 represents control for the first appliance (APPL.1). LED3-LED6 correspond to appliances APPL.2 through APPL.5 based on the x and y

movement of the joystick. Connect the joystick module across CON6 using external jumper wires and connect 230V AC across CON6.

After assembling the circuits on PCB boards, power on the circuit. Observe the power LED1; it should be on. LED2 on the receiving board blinks whenever the joystick is pressed. Test the blinking of the oth-

er LEDs by using joystick movements in different directions (left, right, up, and down). If all LEDs respond correctly to the joystick movements, everything is working as expected. Then connect the appliances to the respective relays and connect 230V AC as well.

**Bonus.** You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/diy-joystick-arduino-based-5-appliances-control> **EFY**

### EFY Note

The source code of this project is available

for free download at [www.electronicsforu.com](http://www.electronicsforu.com)